Practice Set 7

This covers Lectures 24 - 27

(1) Find the Fourier transform of the following signals. You will find the properties discussed in Lec 27 useful.

a. $\text{sinc}^2(t)$

b. $x(t) = \begin{cases} \text{sinc}^2(t) & \text{if } -\frac{1}{2} < t < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$

c. $\exp[-|t-5|]$.

d. $\dfrac{e^{-at}u(t) - e^{-bt}\cdot u(t)}{b-a}$

e. $t \cdot e^{-t^2}$.

f. $t \cdot e^{-|t|}$

g. $\dfrac{4t}{(1+t^2)^2}$ (use part f to answer this)

h. $e^{-3t} \cdot e^{j2\pi t} u(t)$

(2) Assuming $x(t) \xleftrightarrow{F} X(\omega)$, find the Fourier transform of the following:

a. $x(2t-5)$

b. $\mathrm{Re}[x(t)] = \dfrac{x(t) + x^*(t)}{2}$

c. Even part of $x = \dfrac{x(t) + x(-t)}{2}$

d. $x(t) \cdot \cos(\omega_0 t)$ and $x(t) \sin(\omega_0 t)$

e. $x(t) \cos(\omega_0 t + \phi)$ and $x(t) \sin(\omega_0 t + \phi)$.

f. $\dfrac{d^2 x}{dt^2}$, assuming all relevant mathematical conditions are satisfied

Extend to $\dfrac{d^n x}{dt^n}$

g. $t^2 \cdot x(t)$, assuming relevant conditions are satisfied

Extend to $t^n \cdot x(t)$, $n \geq 1$.

h. $(t^2 + 2t + 1)\, x(t)$.

(3) For any signal x, use x^- to denote its "reflection", i.e.,

$$x^-(t) = x(-t)$$

$$X^-(\omega) = X(-\omega).$$

Then the duality result can be stated as follows:

If $X = \mathcal{F}(x)$ then $\mathcal{F}(X) = 2\pi x^{-}$

i.e. $\boxed{\mathcal{F}(\mathcal{F}(x)) = 2\pi \cdot x^{-}.}$

Find: (a) $\mathcal{F}(\mathcal{F}(\mathcal{F}(x)))$

(b) $\mathcal{F}(\mathcal{F}(\mathcal{F}(\mathcal{F}(x))))$

(4) Find the Fourier transform of the following signals:

a. $t \cdot e^{-at} u(t)$, with $a > 0$.

Hint: Use integration by parts in $\int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt$.

b. For part (a): does the Fourier inversion theorem apply?

Answer: Yes; explain why.

c. Can we apply duality to $t \cdot e^{-at} u(t)$, $a > 0$?

What do we obtain via duality?

d. Use part (a) to find the Fourier transform of

$t^2 e^{-at} u(t)$, $a > 0$.

5. Find constants a and b such that

$$a \cdot e^{-bt^2} \xleftrightarrow{F} a \cdot e^{-b\omega^2}.$$

(6) Find the <u>principal value</u> of the following integrals:

a. $\displaystyle\int_{-\infty}^{+\infty} \operatorname{sgn}(t-1)\,dt$

b. $\displaystyle\int_{-\infty}^{+\infty} \sin(\omega_0 t)\,dt$

c. $\displaystyle\int_{-\infty}^{+\infty} \operatorname{sinc}(at)\,dt, \quad a>0.$

(7) <u>About sinc</u>

We want to show that sinc is <u>not</u> absolutely integrable. To do so, use the following hint. Explain why each step in the hint is valid:

$$\int_0^{\infty} \frac{|\sin \pi t|}{\pi t}\,dt = \sum_{n=0}^{\infty} \int_n^{n+1} \frac{|\sin \pi t|}{\pi t}\,dt$$

$$\geq \sum_{n=0}^{\infty} \frac{1}{\pi(n+1)} \int_n^{n+1} |\sin \pi t|\,dt.$$

$$= \left(\int_0^{\pi} |\sin \pi t|\,dt\right) \times \sum_{n=0}^{\infty} \frac{1}{\pi(n+1)}.$$